

$$T1: F(\omega) = \int_{-\infty}^{+\infty} f(t) e^{-i\omega t} dt$$

$$f(t) = \begin{cases} \cos t + \sin t, & |t| \leq \pi \\ 0, & |t| > \pi \end{cases}$$

$$= \int_{-\pi}^{\pi} (\cos t + \sin t) e^{-i\omega t} dt$$

$$= \int_{-\pi}^{\pi} \cos t e^{-i\omega t} dt + \int_{-\pi}^{\pi} \sin t e^{-i\omega t} dt$$

$$T2: \int_{-\pi}^{\pi} \cos t e^{-i\omega t} dt = \int_{-\pi}^{\pi} \cos t \cdot (\cos \omega t - i \sin \omega t) dt$$

$$= \int_{-\pi}^{\pi} \cos t \cos \omega t dt - i \int_{-\pi}^{\pi} \cos t \sin \omega t dt$$

$$= \int_{-\pi}^{\pi} \frac{1}{2} [\cos(t+\omega t) + \cos(t-\omega t)] dt - i \int_{-\pi}^{\pi} \frac{1}{2} [\sin(t+\omega t) - \sin(t-\omega t)] dt$$

$$= \frac{1}{2} \left. \frac{\sin((1+\omega)t)}{1+\omega} \right|_{-\pi}^{\pi} + \frac{1}{2} \left. \frac{\sin((1-\omega)t)}{1-\omega} \right|_{-\pi}^{\pi} - 0 \quad (\text{奇函数})$$

$$= \frac{1}{2} \left[\frac{\sin((1+\omega)\pi)}{1+\omega} + \frac{\sin((1-\omega)\pi)}{1-\omega} \right] + \frac{1}{2} \left[\frac{\sin((1-\omega)\pi)}{1-\omega} + \frac{\sin((1+\omega)\pi)}{1+\omega} \right]$$

$$= \frac{\sin((1+\omega)\pi)}{1+\omega} + \frac{\sin((1-\omega)\pi)}{1-\omega}$$

$$= \frac{(1-\omega) \sin((1+\omega)\pi) + (1+\omega) \sin((1-\omega)\pi)}{1-\omega^2}$$

$$= \frac{(1-\omega)(-\sin \omega \pi) + (1+\omega) \sin \omega \pi}{1-\omega^2} = \frac{\sin \omega \pi (1+\omega + 1-\omega)}{1-\omega^2} = \frac{2 \sin \omega \pi}{1-\omega^2}$$

$$\text{类似: } \int_{-\pi}^{\pi} \sin t e^{-i\omega t} dt$$

$$= \int_{-\pi}^{\pi} \frac{e^{it} + e^{-it}}{2i} e^{-i\omega t} dt = \frac{1}{2i} \int_{-\pi}^{\pi} e^{i(1-\omega)t} + e^{-i(1+\omega)t} dt$$

$$= \frac{1}{2i} \left. \frac{e^{i(1-\omega)t}}{i(1-\omega)} \right|_{-\pi}^{\pi} + \frac{1}{2i} \left. \frac{e^{-i(1+\omega)t}}{-i(1+\omega)} \right|_{-\pi}^{\pi}$$

$$= \frac{1}{2i} \left[\frac{e^{i(1-\omega)\pi}}{i(1-\omega)} - \frac{e^{-i(1-\omega)\pi}}{i(1-\omega)} + \frac{e^{-i(1+\omega)\pi}}{-i(1+\omega)} - \frac{e^{i(1+\omega)\pi}}{-i(1+\omega)} \right]$$

$$= \frac{1}{2i} \left[\frac{\cancel{\cos((1-\omega)\pi)} + i \sin((1-\omega)\pi) - \cancel{\cos((1-\omega)\pi)} + i \sin((1-\omega)\pi)}{1-\omega} - \frac{\cancel{\cos((1+\omega)\pi)} - i \sin((1+\omega)\pi) - \cancel{\cos((1+\omega)\pi)} - i \sin((1+\omega)\pi)}{1+\omega} \right]$$

$$= \frac{1}{2i} \left[\frac{2i \sin((1-\omega)\pi)}{1-\omega} + \frac{2i \sin((1+\omega)\pi)}{1+\omega} \right]$$

$$= \frac{\sin((1-\omega)\pi)}{1-\omega} + \frac{\sin((1+\omega)\pi)}{1+\omega} = \frac{2 \sin \omega \pi}{1-\omega^2}$$